Numerical on Performance Parameters - Week 1

Q1. Suppose a client machine A is communicating with a data center B located 8000 km away from A. How long will it take for a response sent by server to reach client A?(Assume speed of light in cable is 2e8 m/s).

(by default 3x10^8)

2e8 = 2x10^8

```
          —-------request—->>---------
 A                                        B
client  —-<<----- response—-------     server
            8000km
```

Speed = Distance / time time=distance/speed

Latency is the delay between sending a request and receiving a response in a system

Latency= round trip time

= request time+response time+delays

= (8000 x 1000 m/ 2 x 10^8 m/s)+(8000 x 1000 / 2 x 10^8)

= 2 x 8000 x 1000/2 x 10^8 (m/m/s) s

= 8 x 10^-2 s 1 sec= 1000ms

= 0.08s = 0.08 x 1000 ms = 80 ms

Q2 Suppose a client machine C is communicating with a data center D located 15000 km away from C. Assume that the TCP connection has been established and is kept alive. If each new request can be sent only after receiving an acknowledgement from D for the previous request, then what is the maximum number of successful requests that can be sent from C to D in one second? (Assume speed of light in cable is 2e8 m/s).

```
     —-------request—->>---------                    2e8 = 2x10^8
C                                   D
     —-<<----- response—-------
               15000km
```

RTT= 2 x 15000 x 1000 / 2 x 10^8 s
= .15 s (one request-response cycle)
= 150 ms
N = 1000/ 150 (each)
= 6.67
1000 ms = 150+150+150+150+150+150= 900ms 100ms
While 7th request, 75ms+ 25 ms
6 successful request

Q3 Consider a client which is located 6000 km from the server makes a request through the cable. Suddenly after the request reaches the server, the cable breaks and the response is now to be sent to the client via air. This change of medium caused an additional delay of 75 ms at the server end. How long will the client have to wait before receiving the response?(speed on cable = 2e8 m/s and in air 3e8 m/s)

```
     —-------request—->>---------  cable          2e8 = 2x10^8 1s=1000ms
A                                B
     —-<<----- response—-------- air
              6000km
```

Latency = RTT

= tr+ Tr +delays at server end

= (6000 x 1000m/2 x 10^8 m/s)+ (6000 x 1000/ 3 x 10^8) + .075 s (75/1000)

= .03+.02+.075

= .125 s

= 125 ms

Q4. A client C and the server S located, 12000 km apart are connected with the help of a cable. A request was sent by the client C to server S, but while sending back the response to the client, the cable breaks and the response is now sent back via air. This change in medium caused an additional delay of 20 msec. As compared to the Round Trip Time of the healthy network, the Round Trip Time (RTT) of the faulty system would _____.

[The speed of light in air is $3 \, x \, 10^8$ m/s and that on a cable is $2 \, x \, 10^8$ m/s]
[MCQ: 4.5 marks]

A. Increase
B. Decrease
C. Remain Same
D. Insufficient Information

Answer. C

Case1: Healthy network

RTT1 = 2 x 12000 x 1000 / 2 x 10^8 = 120 ms

Case 2: Faulty network

RTT2= (12000 x 1000/ 2 x 10^8) + (12000 x 1000/ 3 x 10^8) + .02s
= .06+ .04 + 0.02
= 0.12s
= 120 ms

Q5. For a network bandwidth of 8Gbps, what should be the size of each request in MB if 5000 such requests are to be sent over the network per second?(use 1KB=1000 Bytes, 1MB=1000KB and so on)

Bits and Bytes

1 Byte =8 bits
1 KB = 1000 Byte
1 MB = 1000 KB = 10^6 Bytes
1 GB = 1000 MB = 10^6 KB = 10^9 Bytes
1 TB = 1000 GB = 10^6 MB = 10^9 KB = 10^12 Bytes

Time conversion
1sec = 1000 milliseconds
1 millisec = 10^-3 s
1 microsecond = 10^-3 millisec = 10^-6 sec
1 nanosecond =10^-3 micros =10^-6 mills= 10^-9 sec

1 million = 10^6 unit
1 billion = 10^3 million = 10^9 unit
1 trillion = 10^3 billion = 10^6 million = 10^12 unit

Bandwidth = (no of requests/s) x size of request
8000Mb/s = 5000 req/s x S
S = 8000/5000
= 1.6 Mb
= 1.6/8 M bytes = 0.2 MB

Bandwidth> b> bits
Storage, size of request> B> bytes
BW = req/s x size of request
BW = data/s

Q6. A certain network connection with a bandwidth of 12 Mbps allows making 4000 requests to a server per second. If each request is modified to have an additional data of 6 Kb, what should be the additional bandwidth required to maintain the rate of 4000 requests per second?

[MCQ: 3 points]

A. 12 Mbps
B. 24 Mbps
C. 36 Mbps
D. 48 Mbps

Answer: Option B

Bandwidth = (no of requests/s) x size of request
S = 12 Mbps / 4000 = 12000 Kbps/4000 = 3 Kb
New S = 3 Kb + 6Kb = 9Kb

New BW = 4000 x 9 Kb
= 36 Mbps

Additional BW required = 36-12= 24 Mbps

Q7. A client sitting in a train is connected to a server through a network tower that has a range of 4.5 kilometers and a constant bandwidth of 5 Mbps. The train is at a negligible distance from the tower and is moving away from it at a speed of 180 kmph. How much data (in Megabytes) will be used by the client before the train completely moves out of the coverage area, assuming that the client is consuming the entire bandwidth?

[MCQ: 3 points]

A. 45 MB
B. 450 MB
C. 5.6 MB
D. 56.25 MB

Answer: Option D

T—----------->----------|----beyond range
<------ 4.5km ----->

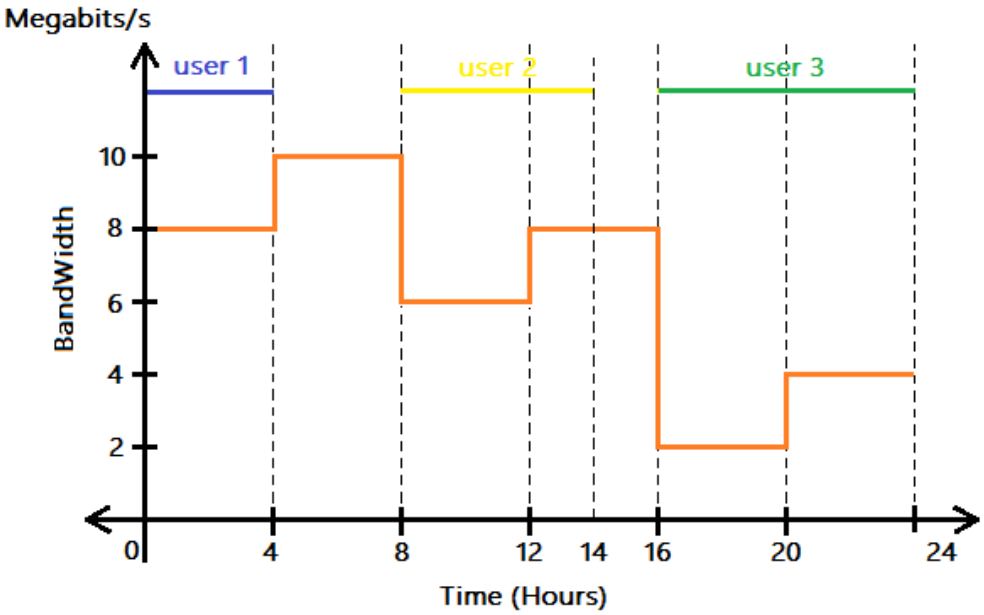
BW= data/s > data = BW x time Mbps = Mb/s
Time = distance/speed

t= 4.5km/180km/h = 4.5 x 3600/180 s =90 s 1 hr = 60 m= 3600s

data = 5Mb/s x 90 s
= 450 Mb
= 450/8
= 56.25 MB

Q8. Consider the following graph that represents the variation in bandwidth of a network for an entire day (24 hours). Three users were connected to the network at three different times of the day. What is the total data consumed in GigaBytes by all the users in 24 hrs?

[MCQ: 4.5 points]



A. 547.2 GB
B. 12 GB
C. 43.2 GB
D. 345.6 GB

Answer: Option C

BW= data/s > data = BW x time

data= ((8 x 4 + 6 x 4 + 8 x 2 + 2 x 4 + 4 x 4)Mb/s x 3600s)/8 MB
= 96 x 3600/8 MB
= 43200 MB
= 43.2 GB

Q9 A certain video on the web occupies 2 megabytes of memory of the server. If there are 5 million concurrent viewers of that video assuming that each viewer requires an individual connection to the server to view the video, what should be the minimum RAM requirement of the server that can process all the viewers simultaneously?

Solution:

	Instances	
Client	**RAM**	**SERVER**
A	video1	video1 (2 MB)
B	video1	

Piyush = 1 client x size
Piyush , vishu = 2 clients x size
Piyush, vishnu, jigyasa = 3 clients x size
N clients = N x size of video
RAM requirement N = N x size of video
RAM req= 5 x 10^6 x 2MB
= 10 x 10^12
= 10 TB

Q10: A server using the inbuilt http module of Python, is running for a directory `My_app` whose file structure and content of each file is given below.

Folder: My_app

```
My_app
   |_ home.html
   |_ first.html
   |_ main.html
```

File: home.html

```
<h1>Hello from Home!</h1>
```

File: first.html

```
<h1>Hello from First!</h1>
```

File: main.html

```
<h1>Hello from Main!</h1>
```

What will be rendered by the browser for the URL: `http://localhost:8000/first.html` assuming that 8000 is the port of connection?

A.

Directory listing for /

- first.html
- home.html
- main.html

B.

Hello from First!

C.

Hello from Home!

D.

Hello from Main!

Answer: Option B